

NeuroSense: Leakage-Aware, Explainable EEG Decoding

An honest look at when EEG machine learning generalizes, and when it only appears to

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Abstract

Machine learning on EEG is easy to evaluate incorrectly. Because EEG is a time series with strong short-range correlation, randomly splitting a single continuous recording into training and test sets lets a model score highly by recognizing near-duplicate neighbors rather than by learning a generalizable mapping. This report studies that failure across two datasets and two paradigms, under a single prespecified model and an evaluation protocol whose unit of inference is the subject or the label block, not the epoch. On the UCI EEG Eye State dataset, a naive random split gives a balanced accuracy of 0.533, a chronological 70/30 holdout gives 0.416 (well below the 0.767 majority baseline), an expanding-window evaluation across label blocks gives 0.380 (95% CI 0.230 to 0.545), and a pre-specified leave-one-merged-block-out (super-block size 4 native blocks, declared before evaluation) gives balanced accuracy 0.456 (95% CI 0.270 to 0.605) and AUC 0.533 (95% CI 0.396 to 0.629); a `DummyClassifier(strategy="prior")` on the same merged splits scores balanced accuracy 0.400 (95% CI 0.200 to 0.500) and AUC 0.500 (95% CI 0.500 to 0.500), so the primary model is within a fold of the dummy. The retained leave-one-native-block-out diagnostic gives pooled AUC 0.437 for the real model and 0.000 for the same dummy on the same splits, which shows that the below-chance pooled AUC is a splitting artefact rather than electrode drift. Every leakage-safe Phase 1 protocol reports its own baselines, and the temporal gap between train and test is chosen from a measured feature-autocorrelation diagnostic. On the PhysioNet Motor Movement/Imagery dataset, within-subject shuffled-trial CV reaches a mean per-subject balanced accuracy of 0.607 (95% CI 0.522 to 0.700) and AUC of 0.632 (95% CI 0.534 to 0.735), with 8 of 10 subjects above 0.5 on both (a descriptive count, not a significance claim). A newly added within-subject leave-one-run-out evaluation across the three imagery runs gives balanced accuracy 0.588 (CI 0.529 to 0.648) and AUC 0.657 (CI 0.598 to 0.718) across 30 folds; it tests transfer across recording runs of the same session but does not establish transfer to a new session. Cross-subject leave-one-subject-out sits at chance (balanced accuracy 0.474, CI 0.432 to 0.516; AUC 0.511, CI 0.447 to 0.583), and per-fold labels, predictions, and probabilities are persisted in the results JSON so that a reader can rescore without re-fitting. Saved logistic-regression coefficients are listed for both phases. In Phase 2 the largest mean absolute weights are on motor-montage feature names. In Phase 1 the merged-block score overlaps a same-split dummy, and an earlier reading of those weights as ocular artifact is removed. The evaluation procedures used here (chronological holdout, walk-forward splits, leave-one-subject-out, leave-one-run-out, CSP plus LDA) are standard. This report does not present them as a new method.

1 Introduction

EEG underpins brain-computer interfaces, attention and fatigue monitoring, and a broad range of neurotechnology. Applying machine learning to EEG, however, requires care that is often skipped in introductory work: the signal is time-dependent, and samples that are close in time are highly

correlated. When such samples are split randomly into training and test sets, a model can appear highly accurate while mostly recognizing repeated temporal patterns it has already seen, rather than learning structure that transfers to new data.

The questions asked here are numerical. On this UCI file, what balanced accuracy does the pre-specified model get under a random split, a chronological holdout, a walk-forward split, and leave-one-merged-block-out, each compared with a same-split dummy? On this 10-subject PhysioNet subset, what balanced accuracy and AUC does the same model get within subject and under leave-one-subject-out? Coefficient vectors are saved with those fits. They are not compared with a physiological template.

A note on provenance. An earlier version of this report presented both phases with separate feature pipelines, different models per phase, and pooled epoch-level metrics. The analysis was subsequently rebuilt so that preprocessing is fitted inside each fold and the reported interval resamples groups. Phase 1 uses delta, theta, alpha, and beta. Phase 2 uses mu and beta. The two feature sets are not the same, and the two datasets are not a controlled comparison. Three claims from that earlier version did not survive the rebuild and are marked as withdrawn where they appear. A follow-up audit in September 2026 added a fold-safe regression guard, a feature-autocorrelation diagnostic that drives the choice of temporal gap, an expanding-window evaluation across label blocks, and per-split baselines. A second audit round replaced the single-class leave-one-block-out primary group protocol with a pre-specified leave-one-merged-block-out, paired every group protocol with a same-splits `DummyClassifier(strategy="prior")` diagnostic, added within-subject leave-one-run-out to Phase 2, persisted per-fold labels and predictions in the Phase 2 JSON, and tightened reporting language throughout: “at chance” is not used without a protocol-specific baseline, “N of 10 above 0.5” is labelled as a descriptive count rather than a significance claim, and the standard deviation across subjects is reported separately from the bootstrap CI of the mean. The audits changed no primary model and no substantive conclusion. Pre-audit results and figures are preserved verbatim under `reports/results/archive/` and `reports/figures/archive/`, and the change list is recorded in `reports/AUDIT_LOG.md`.

2 Data and Methods

Phase 1: UCI EEG Eye State. A single continuous 117-second recording from one subject, 14 channels sampled at 128 Hz, labeled for eyes open versus closed. The label runs in only 24 contiguous blocks with a median duration of about 3.9 seconds, of which 19 contain at least one complete window. Several channels contain single-sample electrode pops reaching roughly 700,000 against a baseline near 4,000; these are clipped per channel, using thresholds fitted on training data only. The signal is segmented into one-second non-overlapping windows, yielding 100 windows and 56 band-power features. Every Phase 1 figure below therefore rests on 100 observations from a single person.

Phase 2: PhysioNet Motor Movement/Imagery. Imagined left versus right fist movement across 10 subjects at 160 Hz, 383 trials in total drawn from runs R04, R08, and R12 (the pre-audit cache had 437 trials; the difference is that an older MNE version retained more boundary-adjacent epochs after concatenation, and the round-2 re-extraction with run IDs preserved uses MNE 1.8’s stricter default). Features are mu (8 to 13 Hz) and beta (13 to 30 Hz) band power over a 13-channel sensorimotor strip, giving 26 features per trial. The imagery-run identifier (R04, R08, or R12) is preserved per trial, so a leave-one-run-out evaluation is possible within subject.

Features. For both phases, band power is computed per channel and window using Welch’s method

and integrated over each band. The delta, theta, alpha, and beta bands are used in Phase 1; the motor-relevant mu and beta bands in Phase 2. The lowest band edge is 1 Hz rather than 0.5 Hz, because a one-second Welch segment resolves frequency only to 1 Hz. The gamma band (30 to 45 Hz) is not used.

Model. A single model is prespecified for both phases and all headline results: logistic regression with $C = 1.0$, balanced class weights, and a fixed random seed. Support vector machines and random forests appear only as secondary comparisons. Fixing one model in advance avoids the failure mode, present in an earlier version of this work, of assembling a headline result from the strongest metric of whichever model produced it.

Evaluation. Chronological holdout, walk-forward evaluation, leave-one-group-out, and CSP plus LDA (Ramoser, Müller-Gerking, and Pfurtscheller, 2000) are standard procedures. They are applied here. They are not proposed as a new method. All preprocessing, including artifact clipping, band-power extraction, and feature scaling, is fitted inside the cross-validation fold. A regression test in the test suite mutates several held-out rows to extreme values, refits the Phase 1 pipeline on the same training rows, and asserts that the fitted clipper thresholds, the fitted scaler statistics, and the transformed training features are bit-identical to a baseline without the mutation. For Phase 1 the grouping unit is the contiguous label block, since the dataset contains one subject; for Phase 2 it is the subject. In both cases one model is fitted per held-out group and that group’s metrics are stored as a single row, so the headline figure is a mean over groups with a 95% percentile bootstrap interval that resamples groups. Pooled epoch-level metrics are reported only as a descriptive appendix: pooling correlated epochs and treating them as independent both understates uncertainty and, for threshold-free metrics, distorts the estimate itself. Every leakage-safe split reports its own baselines (majority-class, stratified dummy, uniform dummy) computed on that split’s test set alone; no baseline is shared across protocols. A naive shuffled split is included in each phase purely as a contrast, to quantify the inflation that leakage produces.

Temporal gap. For Phase 1 the choice of temporal gap between train and test in the chronological and expanding-window protocols is derived from a measurement rather than picked by hand. Mean absolute autocorrelation of the fold-safe band-power features is computed at lags 1 through 5; the recommended gap is the smallest lag at which that value first drops below a fixed threshold of 0.30. On this recording every lag sits in $[0.07, 0.10]$, so the recommendation is 0 windows. The measured mean absolute autocorrelations at lags 1 through 5 are in $[0.07, 0.10]$, all below the preset threshold of 0.30, so the recorded gap is 0 windows. The measurement, the threshold, and the recommendation are in the results JSON.

3 Results

4 Phase 1: Eye-State Detection

Because the label runs in long blocks, neighboring samples are near-identical and share a label. A random split scatters these neighbors across training and test, so a model scores well by recognizing duplicates. Holding everything else fixed and changing only how the split is drawn produces the performance gap shown in Figure 1.

The chronological holdout gives a balanced accuracy of 0.416, which beats a stratified dummy (0.366) by a small margin but sits well below the 0.767 majority-class baseline. Expanding-window across 13 forward-walked label blocks gives 0.380 (0.230 to 0.545). Those folds are single-class, so a stratified-dummy comparison is not reported for them. The pre-specified leave-one-merged-

Phase	Protocol	Bal. accuracy	Macro F1	ROC-AUC
1	Naive random window split	0.533	0.533	0.566
1	Chronological 70/30 holdout (gap = 0)	0.416	0.330	0.491
1	Expanding-window (13 folds, gap = 0)	0.380 [0.230, 0.545]	0.283 [0.165, 0.432]	undefined
1	Leave-one-merged-block-out (5 folds, primary)	0.456 [0.270, 0.605]	0.433 [0.255, 0.582]	0.533 [0.396, 0.629]
1	Same splits, DummyClassifier(prior) diagnostic	0.400 [0.200, 0.500]	0.269 [0.123, 0.374]	0.500 [0.500, 0.500]
1	Leave-one-native-block-out (19 folds, diagnostic)	0.482 [0.334, 0.630]	0.369 [0.239, 0.515]	undefined
2	Naive random trial split	0.504	—	0.521
2	Within-subject shuffled trial CV	0.607 [0.522, 0.700]	0.605 [0.518, 0.698]	0.632 [0.534, 0.735]
2	Within-subject leave-one-run-out (30 folds)	0.588 [0.529, 0.648]	0.556 [0.490, 0.622]	0.657 [0.598, 0.718]
2	Cross-subject (LOSO)	0.474 [0.432, 0.516]	0.427 [0.380, 0.474]	0.511 [0.447, 0.583]

Table 1: All results under the prespecified model. Brackets give 95% percentile bootstrap intervals re-sampling blocks or super-blocks (Phase 1) or subjects (Phase 2). Per-split baselines on the Phase 1 chronological test segment: majority-class 0.767, stratified dummy 0.366, uniform dummy 0.339. On the naive random Phase 1 split the majority baseline is 0.550. Native leave-one-block-out and expanding-window folds are single-class by construction, so their majority-class is trivially 1.0 and the meaningful reference is chance (0.5). Leave-one-merged-block-out uses 4 native blocks per super-block, a rule declared in `src/phase1_eyestate.py` before evaluation; 4 of 5 super-blocks contain both classes so per-fold AUC is defined. Within-subject leave-one-run-out on Phase 2 is added by the 2026-09 audit; see `reports/results/phase2_results.json` for its per-fold rows.

block-out primary protocol gives balanced accuracy 0.456 (0.270 to 0.605) and AUC 0.533 (0.396 to 0.629), while a `DummyClassifier(strategy="prior")` on the same splits gives balanced accuracy 0.400 (0.200 to 0.500) and AUC 0.500 (0.500 to 0.500); the two intervals overlap heavily and the point estimates differ by less than one fold, so this is not evidence of decoding. Leave-one-native-block-out is retained only as a diagnostic (single-class folds by construction, AUC undefined per fold). The naive random window split reaches 0.533. Its stratified dummy on the same rows is 0.552.

The 2026-09 audit added a regression test that fabricates a held-out epoch with extreme values, refits the Phase 1 pipeline on the training epochs, and requires the fitted clipper thresholds, the fitted scaler statistics, and the transformed training features to be bit-identical to a baseline without the mutation. A negative control confirms that this guard is not vacuous: a globally fitted clipper on the mutated data does shift its thresholds. Every Phase 1 fold row also carries a provenance block recording its test-window start-sample indices, block ids, and a hash of the training indices, so a fold in the JSON can be re-located in the raw recording without rerunning the code.

One structural caveat deserves emphasis. Every contiguous native label block is single-class by construction, so a held-out native block (in expanding-window or leave-one-native-block-out) contains only eyes-open or only eyes-closed windows. ROC-AUC cannot be computed on a single-class test set, and balanced accuracy degenerates into the recall of whichever class the block contains. An earlier version of this report gave a leave-one-block-out AUC range of 0.28 to 0.37 and read below-chance AUC as evidence that the features encode recording segment rather than eye state. That reading is removed. Per-fold AUC is undefined because each native block is single-class. The pooled AUC, formed by concatenating predictions from separately trained folds, is 0.437 for the prespecified model and 0.000 for `DummyClassifier(strategy="prior")` on the same splits. The dummy does not use the features. The primary group protocol is now leave-one-merged-block-out with a super-block size of 4 native blocks, declared in `src/phase1_eyestate.py` before evaluation; 4 of 5 super-blocks contain both classes, so per-fold AUC is defined and the same-splits dummy comparison remains as a sanity check. Native LOBO is retained only as a labelled diagnostic.

Phase 1 | UCI EEG Eye State, 100 windows, 1 subject
feature autocorrelation lag1=0.09, lag2=0.09, lag3=0.10, lag4=0.07, lag5=0.08 → gap = 0 windows (already_below)

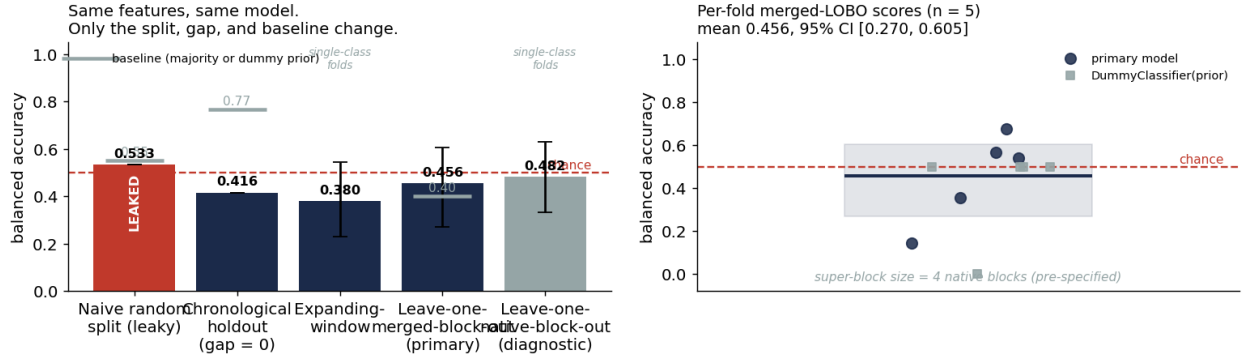


Figure 1: Left: balanced accuracy under five leakage-aware splits of the same features and model, with the chance level marked in red. The majority-class baseline is shown for the two splits where it is informative (naive at 0.55, chronological at 0.77); for leave-one-merged-block-out the DummyClassifier(prior) score on the same splits is shown (0.40); for expanding-window and native leave-one-block-out the caption annotates “single-class folds” because the majority-class baseline is trivially 1.0. The chosen temporal gap of 0 windows is derived from the feature-autocorrelation diagnostic shown in the header. Right: the per-fold merged-LOBO scores that the mean averages over, with both the primary model and the DummyClassifier(prior) plotted so the reader can see the two overlap.

4.1 Coefficients

Signed standardized logistic-regression coefficients are stored for each merged-block training fold. The largest mean coefficient is F4_beta (+1.43). An earlier version of this report read a frontal ranking as ocular artifact. That sentence is removed. No blink annotation and no occipital-alpha contrast are computed in this repository.

The Phase 1 merged-block balanced accuracy is 0.456 (0.270 to 0.605). The same-splits dummy prior balanced accuracy is 0.400 (0.200 to 0.500). The intervals overlap. Both panels of Figure 3 plot signed standardized logistic-regression coefficients from the folds that produced the reported scores. Earlier drafts paired permutation importance on one panel with coefficients from a model fit to all subjects on the other and called those the same method. They were not. On this one recording, under these splits, the prespecified model does not beat the same-split dummy. No claim is made about other published pipelines on the UCI eye-state file.

5 Phase 2: Motor Imagery

The same band-power estimator is applied to trials from 10 subjects (Figure 2). Within a subject, shuffled-trial cross-validation gives a mean per-subject balanced accuracy of 0.607 (0.522 to 0.700) and AUC of 0.632 (0.534 to 0.735). Eight of 10 subjects are above 0.5 on each metric. That count is not a significance test. Both intervals exclude 0.5. Leave-one-run-out trains on two of a subject’s three imagery runs (R04, R08, R12) and tests on the third. Across 30 folds it gives balanced accuracy 0.588 (0.529 to 0.648, 19 of 30 folds above 0.5) and AUC 0.657 (0.598 to 0.718, 25 of 30 folds above 0.5). Shuffled-trial balanced accuracy on the same subjects is 0.607. The three runs come from the same recording, so this comparison does not test a later session. Per-fold LORO

rows and per-fold labels, predictions, and probabilities are stored in the results JSON.

Across subjects, leave-one-subject-out gives balanced accuracy 0.474 (CI 0.432 to 0.516, standard deviation across subjects 0.072) and AUC 0.511 (CI 0.447 to 0.583, std 0.115), both at chance. Standard deviation across subjects is the spread between people; the bootstrap CI is the uncertainty of that mean, computed by resampling subjects (not trials). Individual subjects range from 0.353 to 0.600, a spread that a single pooled figure conceals entirely. Every held-out subject's `y_true`, `y_pred`, and `y_proba` are persisted in the JSON so the reader can rescore or diagnose without re-fitting.

An earlier version reported cross-subject AUC as 0.48 and described it as below chance. That figure pooled every held-out subject's predicted probabilities into one array before scoring. The mean of the per-subject AUCs is 0.511 (0.447 to 0.583). The pooled number is kept in the JSON under a descriptive label. The reason the pooled number differs from the subject mean is not identified here.

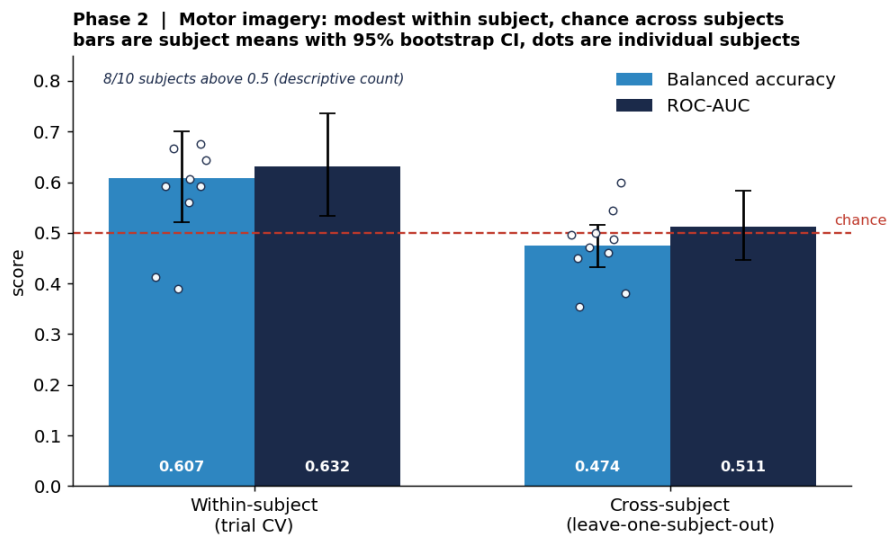


Figure 2: Within-subject decoding is modestly above chance on AUC; cross-subject transfer sits at chance. Bars are subject means with 95% bootstrap intervals; dots are the ten individual subjects.

The five largest mean absolute coefficients from the within-subject fits are `C6_beta`, `C1_mu`, `C3_beta`, `C5_mu`, and `C4_mu` (Figure 3, right). Those names are channels in the 13-electrode montage used to build the features. An earlier sentence said this ranking confirms known physiology. That sentence is removed. No test compares the weight vector with a physiological template.

Leave-one-subject-out balanced accuracy is 0.474 (0.432 to 0.516). The interval includes 0.5. This repository does not fit a subject-identity classifier, so it does not say why the cross-subject score is at that level.

5.1 Standard motor-imagery baseline (CSP + LDA, round 4 extension)

Round 4 of the audit adds Common Spatial Patterns with Linear Discriminant Analysis (CSP + LDA) as a *separately labelled* baseline for Phase 2. The audited band-power result above remains the primary; the CSP baseline is reported here as the standard motor-imagery baseline on the same 10 subjects, the same eligible trials (imagery runs R04, R08, R12), and the

same evaluation splits, so the two are directly comparable. CSP (6 components, log-variance) and LDA are fitted inside training folds only; no learned transformation crosses a fold boundary. Signal filtering is a 4th-order Butterworth zero-phase bandpass 8-30 Hz, applied per epoch on the 13-channel motor strip over a 0.5-3.5 s window. Full settings and per-fold results are in `reports/results/phase2_csp_lda_results.json`.

Within subject, CSP plus LDA balanced accuracy is 0.655 [0.543, 0.773] on shuffled trial CV and 0.640 [0.577, 0.705] on leave-one-run-out, against 0.607 [0.522, 0.700] and 0.588 [0.529, 0.648] for band-power logistic regression on the same splits. Across subjects, CSP plus LDA balanced accuracy is 0.541 [0.478, 0.618] and AUC is 0.654 [0.556, 0.767]. The balanced-accuracy interval includes 0.5. Three of ten held-out subjects are above 0.5 on balanced accuracy, and nine of ten are above 0.5 on AUC. Those are counts, not a test.

Same attribution method both sides: signed standardized coefficients collected inside each evaluation fold. White dots = per-fold values (spread you would otherwise not see).

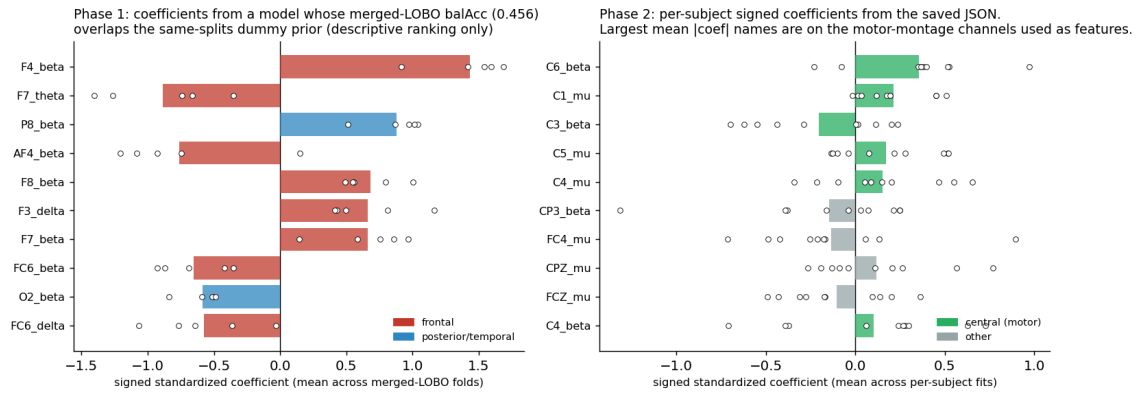


Figure 3: Signed standardized logistic-regression coefficients from the saved evaluation folds. Bars are means. White dots are per-fold values. Right: Phase 2 within-subject fits. The largest mean absolute weights are C6_beta, C1_mu, C3_beta, C5_mu, and C4_mu, which are names in the motor montage used as features. Left: Phase 1 merged-block fits. That model’s balanced accuracy is 0.456 and the same-splits dummy prior is 0.400.

6 Discussion

On the UCI file the random-split balanced accuracy is 0.533 and the chronological holdout is 0.416, against a chronological majority baseline of 0.767. On the PhysioNet subset the random-split balanced accuracy is 0.504 and leave-one-subject-out is 0.474. The gap between a random split and a blocked split is large in Phase 1 and small in Phase 2. Leave-one-subject-out band-power balanced accuracy on these 10 subjects includes 0.5.

Phase 1 and Phase 2 use different hardware, channel counts, sampling rates, and label structures. A difference between their scores is not a comparison of paradigms.

The stored coefficients are the weights of the fitted logistic regressions. Where the Phase 1 score overlaps the dummy, those weights are still the weights. They are not used here as evidence about artifacts or about cortex.

7 Limitations

Phase 1 is a single subject and single session recorded on consumer-grade hardware, so its results are not subject-generalizable and its high-frequency content is unreliable. Windowing produces only 100 windows, which is why every Phase 1 interval is wide, and its single-class label blocks make leave-one-block-out only partially interpretable. Phase 2 uses 10 of the 109 available subjects, so its intervals are wider than necessary. The two phases also differ in hardware, channel count, and sampling rate, so any comparison between them confounds paradigm with recording setup; the contrast between them is offered as a hypothesis rather than a controlled result.

8 Future Work

Round 4 added a CSP + LDA baseline as a separately labelled Phase 2 extension. The natural next steps beyond that are to scale Phase 2 to all 109 available subjects (this report uses 10) to tighten the estimates, and to attempt subject-adaptive transfer, both unsupervised alignment and small-sample calibration, to lift cross-subject decoding above chance under a strict reading of the balanced-accuracy CI. Explanation-faithfulness checks (permuting the top-ranked features and measuring the held-out score drop) and model-randomisation tests would let the attribution ranking be read as physiological evidence rather than as a descriptive summary of one classifier’s weights. A third phase on mental arithmetic is planned, with its hypotheses, ablations, and primary endpoint fixed in advance in a committed preregistration. Deep models (EEGNet, transformers) are deferred until these leakage-safe baselines are established; they are not in scope for the current audit.

Reproducibility

Two documented reproduction paths. Path A reruns from the committed feature cache (`data/processed/physionet_features.npz`) and skips the EDF download; it produces the audited band-power results and rebuilds the three figures and the two markdown tables from the JSONs. Path B rebuilds the cache from raw EDFs via `scripts/download_data.sh` and `python -m src.physionet_features`, and additionally runs the round-4 CSP+LDA extension (`python -m src.phase2_csp_lda`), which needs the raw epochs. The download script hard-fails on any missing or truncated EDF (below 500 KB per file); the feature extractor validates that all 10 expected subjects and all 3 imagery runs per subject (R04, R08, R12) are present, asserts the sampling rate on every EDF equals 160 Hz, and writes `data/processed/physionet_features_manifest.json` with per-subject SHA-256 checksums, per-trial identifiers, channel order, frequency bands, epoch timing, and Welch parameters, so Path A and Path B can be checked against each other byte for byte. The Phase 2 cache used in this report has 383 trials (post-audit re-extraction with MNE 1.8); an older MNE version retained more boundary-adjacent epochs after `mne.concatenate_raws` and yielded 437 trials in the pre-audit build. Point estimates on the trimmed cache are close to the pre-audit numbers and do not change any conclusion. Supported Python is 3.9 (audit ran on 3.9.6); MNE is only required for Path B. Round 4 added a CI workflow (`.github/workflows/tests.yml`) that runs the 73-test suite and rebuilds the figures and tables from JSON on every push and pull request, without downloading the raw dataset.

References

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